

Week 08

Portable Document Format

Web Data Science

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Tuesday

Agenda

- 00:00 – 00:05 → Daily Note check-in
- 00:05 – 00:65 → Notebook lecture & exercises
- 00:70 – 00:75 → Next Class

Daily Note questions

- More heads up before guest speakers to prepare questions
 - Will do!

Next class

- Daily note: <https://bit.ly/info4871f24note>
- Today's number is... 2003
- Working session on Thursday to finalize Module Assignment 02

Thursday

Agenda

- 00:00 – 00:05 → Daily Note check-in
- 00:05 – 01:10 → Show and tell and MA02 working session
- 01:10 – 01:15 → Daily note

Daily Note questions

- Use PyPDF for Module Assignment 3?
 - You definitely can, but it's probably doing too much for a class project
- Use OpenAI cookbook for MA2?
 - Doesn't make much sense for MA2, worth considering for MA3 and Final Project

Module Assignment 02

- Scrape out biographical information about Colorado legislators
 - The Colorado General Assembly consists of 35 Senators and 65 Representatives
 - <https://leg.colorado.gov/legislators/>
 - Scrape out biographical information about all 100 members into a structured format (JSON, CSV, etc.)
 - Name, occupation, party, committee assignments, mailing address, capitol phone, email, district number
 - Do some kind of exploratory data analysis with your data
 - **Cannot** use the legislative directory: https://leg.colorado.gov/sites/default/files/member_directory_032024.xlsx
- Use professional style ([docs](#), [STSI](#), etc.) with good Markdown, sectioning, and links to resources
 - Include documentation and commenting of what each section does
 - Abstract into functions where possible (see Week 5)
 - Use [exception handling](#) to deal with weird cases
 - Do not include giant chunks of HTML output in your notebook
 - Follow norms like using [sleep](#) (see Week 5) to space out requests
- Submit the Notebook (as HTML) and data file to Canvas by **Friday, October 25 at 11:59pm**
- Extra credit: one extra point for every previous legislature, 2016-2023.

Module Assignment 03

- Scrape out other data about the Colorado legislature for your final project
 - Committees, bills, previous legislative sessions from General Assembly website
 - Data from other web sources
 - Examples: [Ballotpedia](#), [fundraising disclosures](#), [election results](#), [Census](#), social media, *etc.*
 - Make your own data
 - Classify legislators' race, gender, *etc.*
- Use professional style (commenting, *etc.*) and follow good web scraping norms in code
- Submit the Notebook (as HTML) and data file to Canvas by **Friday, November 22 at 11:59pm**

Next class

- Daily note: <https://bit.ly/info4871f24note>
- Today's number is...
- New module starting next week: working with APIs and Wikipedia data via API